

SEO Optimization Report

Technical & On-Page SEO for monsun.io

Status: Deployed

Canonical: monsun.io

GSC: Submitted

Site	https://monsun.io
Brand	Monsun · Metocean Intelligence & Consultancy
Report date	5 August 2026
Scope	Technical SEO, on-page structure, content depth, discovery assets
Deploy branch	dev → GitHub Pages (main)

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1 Executive summary

This report documents the SEO optimization work completed for **Monsun** on monsun.io. The site was a Vue 3 + Vite client-side SPA with almost no SEO infrastructure. After the optimization pass, the live site ships crawlable HTML, keyword-aligned copy, discovery files, structured data, and expanded metocean content.

Verdict: Technical and on-page SEO foundations are **deployed and verified** on production. Google Search Console has already been submitted; discoverability in organic search may still take days to weeks while Google crawls and indexes the new pages.

Primary keyword targets

- metocean intelligence
- metocean consultancy / consulting / services
- sea state monitoring, marine forecasting, offshore metocean
- Related long-tail: AI ocean forecasting, offshore wind metocean, satellite ocean data

2 Baseline (before)

Area	Before	Impact
Crawlability	Client-only SPA shell	Crawlers saw empty #app
Title / meta	Monsun only	Weak SERP & social previews
Discovery	No robots / sitemap	Hard for Google to discover URLs
Structured data	None	No rich Organization / Article signals
Headings	Duplicate H1s	Diluted topical focus
Hero copy	Generic “data-driven insights”	Missed primary keywords
Blog depth	80–100 words / post	Thin content risk
Images	Broken /blog-images, 1.9MB hero	Weak UX + LCP risk
Domain	Custom domain live	CNAME file missing in repo deploys

Table 1: SEO gaps identified before the optimization pass.

3 Work completed

3.1 Crawlability & domain

- Static **prerender** of /, /blog, and all blog post URLs at build time (scripts/prerender.mjs)
- GitHub Pages SPA fallback via 404.html
- Canonical origin locked to https://monsun.io
- public/CNAME with monsun.io so orphan main deploys keep the custom domain binding
- Build pipeline: npm run build = Vite build + prerender + sitemap generation

3.2 Technical head & discovery

Asset	Implementation
Titles	Home / blog / posts with Monsun + metocean framing
Meta description	Unique 150-160 char descriptions per route
Canonical	https://monsun.io/... per page
Open Graph / Twitter	title, description, url, image (og-image.jpg)
robots.txt	Allow all · points to sitemap
sitemap.xml	Generated at build from routes + blog markdown
JSON-LD	Organization, WebSite, ProfessionalService, BlogPosting, BreadcrumbList
Favicon	Correct image/x-icon + apple-touch-icon
Runtime SEO	@unhead/vue + useSeo for client route updates

Table 2: Technical SEO assets now present on production.

3.3 On-page structure & brand

- Removed duplicate Header/H1 on home (logo is brand text, not H1)
- Hero H1: **Metocean Intelligence for Safer Offshore Decisions**
- Supporting copy names Monsun + metocean consultancy
- Services / About rewritten around metocean consultancy & intelligence
- Footer email aligned to contact@monsun.io
- Labs / Forecast links updated to https://labs.monsun.io and https://forecast.monsun.io
- Internal links: sitemap → /sitemap.xml, services CTAs → contact

3.4 Content depth

Expanded existing posts and added three pillar articles:

URL path	Role
/blog/what-is-metocean-consultancy	Pillar · commercial intent
/blog/metocean-intelligence-for-offshore-operations	Pillar · category intent
/blog/sea-state-monitoring-best-practices	Pillar · capability intent
/blog/harnessing-ocean-data-with-ai	Expanded · AI / forecasting
/blog/open-data-coastal-management	Expanded · coastal / open data
/blog/offshore-wind-forecasting	Expanded · offshore wind
/blog/satellite-data-integration	Expanded · satellite metocean

Table 3: Blog inventory after content expansion.

Frontmatter now includes description (and related fields) for Unhead + Article schema.

3.5 Performance & media

- Compressed hero image (50KB JPEG)
- Added /public/blog-images/* covers (paths resolve)
- OG share image at /og-image.jpg
- Lazy-loading on below-fold blog cards

4 Live verification

Checked after deploy to production:

Check	Result	Evidence
Homepage title	Pass	Monsun Metocean Intelligence & Consultancy
Prerendered H1 in HTML	Pass	Visible without JS execution
JSON-LD present	Pass	Organization / WebSite / ProfessionalService
/robots.txt	Pass	HTTP 200 · sitemap reference
/sitemap.xml	Pass	HTTP 200 · all key URLs listed
site:monsun.io on Google	Pending	Awaiting crawl/index after GSC submit
“monsun metocean” SERP	Pending	Competitors only — normal until indexed + ranked
Google Search Console	Done	Property + sitemap submitted by team

Table 4: Production verification status as of report date.

5 Sitemap update behaviour

The sitemap is **build-time generated**, not a runtime API.

1. New blog post (.md under `src/blog/`)
2. Push to dev → GitHub Actions build
3. `scripts/prerender.mjs` rediscovers slugs and rewrites `sitemap.xml`
4. Deploy ships the updated file to `https://monsun.io/sitemap.xml`

Without rebuild/deploy, the live sitemap does not change automatically.

6 Google Search Console & discoverability

Search Console for `monsun.io` has already been submitted (property + sitemap). That does **not** mean pages appear in Google search immediately.

Typical timeline:

1. **Crawl** — Googlebot fetches URLs from the sitemap / links
2. **Index** — pages enter Google's index (`site:monsun.io` starts showing results)
3. **Rank** — pages compete for queries such as “monsun metocean” and broader metocean terms

Expectation: Indexing often takes **days to a few weeks**. Ranking for competitive metocean terms takes longer and depends on content freshness, internal links, and external mentions. Re-check periodically with `site:monsun.io` and Search Console **Cov-erage / Pages** reports.

Monitor in Search Console: sitemap status, indexing issues, Core Web Vitals, and which queries begin to impress after indexation.

7 Key repository files

Path	Role
scripts/prerender.mjs	Static HTML prerender + sitemap + 404
public/CNAME	Persist monsun.io on Pages deploy
public/robots.txt	Crawl policy + sitemap pointer
src/config/site.js	Site URL / defaults
src/composables/useSeo.js	Client-side head management
src/seo/schemas.js	JSON-LD helpers
src/blog/*.md	Content + SEO frontmatter
index.html	Baseline meta / OG bootstrap

Table 5: Primary implementation touchpoints.

8 Recommendations for future improvements

Prioritized next steps beyond the current foundation:

8.1 Content & topical authority

- Publish 1–2 new metocean articles per month targeting long-tail queries (e.g. monsoon wave forecasting, port operability, Indonesian / SEA offshore case studies)
- Add author/expert bios and “last updated” dates on pillar posts
- Build a simple topic cluster: each pillar post links to related services + sibling posts with natural anchor text
- Create dedicated landing copy for high-intent phrases if traffic justifies it (e.g. standalone metocean consultancy page)

8.2 International & regional reach

- Add multilingual routes (`id`, `zh`, and other SEA locales) with `hreflang` once English rankings stabilize
- Localize pillar posts first (not only UI chrome) so regional SERPs can rank native-language pages

8.3 Technical evolution

- Consider SSG/SSR (Astro or Nuxt) if content volume and team workflow grow — cleaner HTML, simpler i18n, stronger Core Web Vitals control
- Convert remaining large assets to WebP/AVIF with `srcset`; keep measuring LCP on mobile
- Add FAQ schema on consultancy / services sections where Q&A content exists
- Ensure every new deploy still ships `CNAME` so GitHub Pages custom-domain settings are not wiped

8.4 Measurement & iteration

- Review Search Console weekly for 4–6 weeks after submission: indexed page count, sitemap errors, query impressions

- Track branded vs non-branded queries (“Monsun” vs “metocean consultancy”)
- Set lightweight goals in analytics (contact form submissions from organic sessions)

8.5 Authority & distribution

- Share pillar posts via LinkedIn ([monsunic](#)), partner pages, and Monsun Labs / Forecast with contextual links back to [monsun.io](#)
- Pursue industry citations (guest notes, conference pages, open-data acknowledgements) rather than generic directory spam
- Keep Labs ([labs.monsun.io](#)) and Forecast ([forecast.monsun.io](#)) interlinked with clear brand + keyword anchors

9 Conclusion

Monsun’s public site now has a solid SEO foundation: crawlable pages, correct [monsun.io](#) canonicals, discovery files, structured data, keyword-led homepage, and deeper metocean content. Search Console is already submitted — the next phase is waiting for crawl/index, then compounding with content, measurement, and authority building.

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